

TAO NI (TONY)

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EMPLOYMENT

- | | |
|---|-----------------------|
| King Abdullah University of Science and Technology (KAUST), Saudi Arabia | Oct. 2025 – Present |
| ♦ Assistant Professor in Computer Science, CEMSE Division | |
| City University of Hong Kong, Hong Kong | June 2024 – Sep. 2025 |
| ♦ Postdoctoral Research Fellow | |

EDUCATION

- | | |
|--|-----------------------|
| City University of Hong Kong, Hong Kong | Feb. 2021 – June 2024 |
| ♦ Doctor of Philosophy (Ph.D.) in Computer Science | |
| Australian National University, Canberra, Australia | Feb. 2019 – Dec. 2020 |
| ♦ Master of Computing in Computer Science | |
| Shanghai Jiao Tong University, Shanghai, China | Sep. 2014 – June 2018 |
| ♦ Bachelor of Engineering in Electrical Engineering | |

HONORS/AWARDS

- Best Presentation Award, AIIoTSys Rising Star Forum, 2025
- AIIoTSys Rising Star, 2025
- Outstanding Research Thesis Award, City University of Hong Kong, 2025
- Cybersecurity Best Practical Paper Award, 2024
- People's Choice ASSET Poster Award, 2024
- ACM MobiSys Rising Star, 2024
- ACM MobiSys Student Travel Grant Award, 2024
- Research Tuition Scholarship (RTS), City University of Hong Kong, 2023–2024
- Outstanding Academic Performance Award (OAPA), City University of Hong Kong, 2023–2024
- UbiComp/ISWC CPD Workshop, Best Paper Award, 2021
- City University of Hong Kong Postgraduate Scholarship, 2021–2025

PUBLICATIONS

Refereed Conference Publications (* means co-first authors)

- [C1] **Penetrating the Reality Wall: Long-Range and Through-Wall AR/VR Privacy Eavesdropping via Wireless Power Side Channel**
Tao Ni*, Zehua Sun*, Yihe Zhou, Yuefeng Du, Qingchuan Zhao, Cong Wang
In Proc. of the 33rd ACM Conference on Computer and Communications Security (CCS), 2026.

- [C2] **When VR Meets BCI: (Un)Observable Brainwave-aware Privacy Reconstruction in the Metaverse via Unrestricted Inbuilt Motion Sensors**
Tao Ni, Zehua Sun, Qingchuan Zhao, Wei-Bin Lee, Cong Wang
In Proc. of the 47th IEEE Symposium on Security and Privacy (S&P, Oakland), 2026.
- [C3] **Adversarial Patch EXterminator: Zero-Shot and Patch-Agnostic Defense Framework Against Adversarial Patch Attacks**
 Jiayimei Wang, **Tao Ni**, Guowen Xu, Qingchuan Zhao, Cong Wang
In Proc. of the 35th USENIX Security Symposium (USENIX Security), 2026.
- [C4] **Non-intrusive and Unconstrained Keystroke Inference in VR Platforms via Infrared Side Channel**
Tao Ni, Yuefeng Du, Qingchuan Zhao, Cong Wang
In Proc. of the Network and Distributed System Security Symposium (NDSS), 2025.
- [C5] **SpaceSched: A Constellation-Wide Scheduling System for Resolving Ground Track Congestion in Remote Sensing**
 Zehua Sun, **Tao Ni**, Pengfei Hu, Tao Gu, Weitao Xu
In Proc. of the 31st Annual International Conference On Mobile Computing And Networking (MobiCom), 2025.
- [C6] **RingByte: Enhancing Text-Entry Practicality via A Singular Wearable Rotating Smart Ring**
 Rucheng Wu, **Tao Ni**, Zehua Sun, Jiande Sun, Weitao Xu
In Proc. of the 38th ACM Symposium on User Interface Software and Technology (UIST), 2025.
- [C7] **Omni-Angle Assault: An Invisible and Powerful Physical Adversarial Attack on Face Recognition**
 Shuai Yuan, Hongwei Li, Rui Zhang, Hangcheng Cao, Wenbo Jiang, **Tao Ni**, Wenshu Fan, Qingchuan Zhao, Yuguang Fang
In Proc. of the 42nd International Conference on Machine Learning (ICML), 2025.
- [C8] **The Fluorescent Veil: A Stealthy and Effective Physical Adversarial Patch Against Traffic Sign Recognition**
 Shuai Yuan, Guowen Xu, Hongwei Li, Xingshuo Han, Wenbo Jiang, **Tao Ni**, Qingchuan Zhao, Yuguang Fang
In Proc. of the 39th Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.
- [C9] **RF-Egg: An RF Solution for Fine-Grained Multi-Target and Multi-Task Egg Incubation Sensing**
 Zehua Sun, **Tao Ni**, Yongliang Chen, Di Duan, Kai Liu, Weitao Xu
In Proc. of the 30th Annual International Conference on Mobile Computing and Networking (MobiCom), 2024.
- [C10] **REHSense: Towards Battery-Free Wireless Sensing via Radio Frequency Energy Harvesting**
Tao Ni, Zehua Sun, Mingda Han, Yaxiong Xie, Guohao Lan, Zhenjiang Li, Tao Gu, Weitao Xu
In Proc. of the 25th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc), 2024.
- [C11] **F²Key: Dynamically Converting Your Face into a Private Key Based on COTS Headphones for Reliable Voice Interaction**
 Di Duan, Zehua Sun, **Tao Ni**, Shuaicheng Li, Xiaohua Jia, Weitao Xu, Tianxing Li
In Proc. of the 22nd ACM International Conference on Mobile Systems, Applications, and Services (MobiSys), 2024.
- [C12] **Recovering Fingerprints from In-Display Fingerprint Sensors via Electromagnetic Side Channel**
Tao Ni, Xiaokuan Zhang, Qingchuan Zhao
In Proc. of the 30th ACM Conference on Computer and Communications Security (CCS), 2023.
- [C13] **Uncovering User Interactions on Smartphones via Contactless Wireless Charging Side Channels**
Tao Ni, Xiaokuan Zhang, Chaoshun Zuo, Jianfeng Li, Zhenyu Yan, Wubing Wang, Weitao Xu, Xiapu Luo, Qingchuan Zhao
In Proc. of the 44th IEEE Symposium on Security and Privacy (S&P, Oakland), 2023.
- [C14] **Eavesdropping Mobile App Activity via Radio-Frequency Energy Harvesting**
Tao Ni, Guohao Lan, Jia Wang, Qingchuan Zhao, Weitao Xu
In Proc. of the 32nd USENIX Security Symposium (USENIX Security), 2023.
- [C15] **Exploiting Contactless Side Channels in Wireless Charging Power Banks for User Privacy Inference via Few-shot Learning**

Tao Ni, Jianfeng Li, Xiaokuan Zhang, Chaoshun Zuo, Wubing Wang, Weitao Xu, Xiapu Luo, Qingchuan Zhao
In Proc. of the 29th Annual International Conference on Mobile Computing and Networking (MobiCom), 2023.

[C16] **XPorter: A Study of the Multi-Port Charger Security on Privacy Leakage and Voice Injection**

Tao Ni, Yongliang Chen, Weitao Xu, Lei Xue, Qingchuan Zhao
In Proc. of the 29th Annual International Conference on Mobile Computing and Networking (MobiCom), 2023.

[C17] **FLoRa: Energy-Efficient, Reliable, and Beamforming-Assisted Over-The-Air Firmware Update in LoRa Networks**

Zehua Sun, **Tao Ni**, Huanqi Yang, Kai Liu, Yu Zhang, Tao Gu, Weitao Xu
In Proc. of the 22nd International Conference on Information Processing in Sensor Networks (IPSN), 2023.

[C18] **MobileKey: A Fast and Robust Key Generation System for Mobile Devices**

Keqi Song, Zimeng Zhu, Huanqi Yang, **Tao Ni**, Weitao Xu
In Proc. of the 5th Workshop on Combining Physical and Data-Driven Knowledge in Ubiquitous Computing, ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp CPD Workshop), 2022.

[C19] **A Simple and Fast Human Activity Recognition System Using Radio Frequency Energy Harvesting**

Tao Ni, Yongliang Chen, Keqi Song, Weitao Xu
In Proc. of the 4th Workshop on Combining Physical and Data-Driven Knowledge in Ubiquitous Computing, ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp CPD Workshop), 2021.

Refereed Journal Publications

[J1] **SwiftChannel: Algorithm-Hardware Co-Design for Deep Learning-Based 5G Channel Estimation**

Shengzhe Lyu, Yuhan She, Di Duan, **Tao Ni**, Yu Hin Chan, Chengwen Luo, Ray C. C. Cheung, Weitao Xu
IEEE Transactions on Mobile Computing (TMC), 2025.

[J2] **Characterizing Contactless Side-channel Eavesdropping on Wireless Chargers**

Tao Ni, Chaoshun Zuo, Jianfeng Li, Wubing Wang, Weitao Xu, Xiapu Luo, Qingchuan Zhao
IEEE Transactions on Dependable and Secure Computing (TDSC), 2025.

[J3] **When Good Becomes Evil: Exploring Crosstalk Attack Surfaces on Multi-Port USB Chargers**

Tao Ni, Zehua Sun, Yongliang Chen, Yihe Zhou, Jiayimei Wang, Weitao Xu, Qingchuan Zhao, Cong Wang
IEEE Transactions on Mobile Computing (TMC), 2025.

[J4] **No Trespassing: Ground-view Adversarial Patches for Privacy-aware Management in COTS Robot Vacuum Cleaner**

Shuai Yuan, Guowen Xu, Hongwei Li, Rui Zhang, Hangcheng Cao, Xinyuan Qian, **Tao Ni**, Qingchuan Zhao, Yuguang Fang
IEEE Transactions on Dependable and Secure Computing (TDSC), 2025.

[J5] **A Contemporary Survey of Large Language Model Assisted Program Analysis**

Jiayimei Wang, **Tao Ni**, Wei-Bin Lee, Qingchuan Zhao
Scilight Transactions on Artificial Intelligence (TAI), 2025.

[J6] **A Survey on Backdoor Threats in Large Language Models (LLMs): Attacks, Defenses, and Evaluations**

Yihe Zhou, **Tao Ni**, Wei-Bin Lee, Qingchuan Zhao
Scilight Transactions on Artificial Intelligence (TAI), 2025.

[J7] **FLoRa+: Energy-Efficient, Reliable, Beamforming-Assisted, and Secure Over-The-Air Firmware Update in LoRa Networks**

Zehua Sun, **Tao Ni**, Huanqi Yang, Kai Liu, Yu Zhang, Tao Gu, Weitao Xu
ACM Transactions on Sensor Networks (TOSN), 2024.

[J8] **Emma: An Accurate, Efficient, and Multi-modality Strategy for Autonomous Vehicle Angle Prediction**

Keqi Song, **Tao Ni**, Linqi Song, Weitao Xu
Intelligent and Converged Networks, 2023.

- [J9] **mmSign: mmWave-based Few-Shot Online Handwritten Signature Verification**
Mingda Han, Huanqi Yang, Tao Ni, Di Duan, Mengzhe Ruan, Yongliang Chen, Jia Zhang, Weitao Xu
ACM Transactions on Sensor Networks (TOSN), 2023.
- [J10] **SwipePass: Acoustic-based Second-factor User Authentication for Smartphones**
Yongliang Chen, Tao Ni, Weitao Xu, Tao Gu
ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (UbiComp/IMWUT), 2022.

PROFESSIONAL SERVICE

Conference Organizer & Sponsorship

- ♦ Gold Sponsor, ACM/IEEE International Symposium on Computer Architecture (ISCA), 2026
- ♦ Silver Sponsor, ACM SIGOPS European Conference on Computer Systems (EuroSys), 2026
- ♦ Open Access Sponsor, USENIX Security Symposium (USENIX Security), 2026
- ♦ Session Chair, AAMC International Conference on Mobile Systems, Applications, and Services (MobiSys), 2026
- ♦ Session Chair, ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc), 2024
- ♦ Publicity Chair, International Conference on Application Intelligence and Blockchain Security (AIBlock), 2026

Technical Program Committee (TPC) & Reviewer

- ♦ USENIX Security Symposium (USENIX Security), 2026
- ♦ ISOC Network and Distributed System Security Symposium (NDSS), 2026
- ♦ International Symposium on Research in Attacks, Intrusions and Defenses (RAID), 2026
- ♦ Annual Computer Security Applications Conference (ACSAC), 2026
- ♦ ACM International Conference on Mobile Systems, Applications, and Services (MobiSys), 2026
- ♦ International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous), 2026
- ♦ International Conference on Learning Representations (ICLR), 2025–2026
- ♦ AAAI Conference on Artificial Intelligence (AAAI), 2025–2026
- ♦ Annual Conference on Neural Information Processing Systems (NeurIPS), 2024–2025
- ♦ International Conference on Machine Learning (ICML), 2024–2025
- ♦ Annual Conference on Artificial Intelligence and Statistics (AISTATS), 2024–2026
- ♦ European Conference on Computer Vision (ECCV), 2026
- ♦ ACM International Conference on Multimedia (MM), 2025
- ♦ EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous), 2026
- ♦ IEEE International Conference on Parallel and Distributed Systems (ICPADS), 2025.
- ♦ IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), 2024, 2026
- ♦ IEEE International Conference on Mobility, Sensing and Networking (MASS), 2024

Journal Editorial Board

- ♦ IEEE Transactions on Services Computing (TSC), Associate Editor, 2025–Present
- ♦ Springer Journal of Cybersecurity, Guest Editor, 2024–Present
- ♦ ScienceDirect Journal of Information and Intelligence, Editor, 2024–Present

Journal Reviewer

- ♦ IEEE Transactions on Dependable and Secure Computing (TDSC)
- ♦ IEEE Transactions on Mobile Computing (TMC)
- ♦ IEEE Transactions on Information Forensics and Security (TIFS)
- ♦ IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- ♦ IEEE Transactions on Network and Service Management (TNSM)
- ♦ IEEE Internet of Things Journal (IoTJ)
- ♦ ACM Transactions on Privacy and Security (TOPS)
- ♦ ACM Transactions on Autonomous and Adaptive Systems (TAAS)
- ♦ ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)
- ♦ ACM Transactions on Sensor Networks (TOSN), Distinguished Reviewer